Magnetohydrodynamics: Plasma Confinement and Stability

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Abstract

Magnetohydrodynamics (MHD) describes the macroscopic behavior of electrically conducting fluids, treating plasma as a single conducting fluid coupled to electromagnetic fields. This report presents computational analysis of fundamental MHD wave modes, stability boundaries, and magnetic reconnection dynamics. We solve the MHD dispersion relation for Alfvén and magnetosonic waves, compute growth rates for kink and sausage instabilities in cylindrical plasmas, and model Sweet-Parker magnetic reconnection. The analysis demonstrates how magnetic field tension stabilizes plasma columns, how plasma beta determines wave propagation characteristics, and how resistivity controls reconnection rates. Results include Alfvén velocities of 500–5000 km/s for fusion-relevant parameters, instability growth times of 10–100 μs for laboratory plasmas, and reconnection timescales of milliseconds to seconds depending on magnetic Reynolds number.

1 Introduction

Magnetohydrodynamics unifies fluid dynamics with Maxwell's equations by treating plasma as a single conducting fluid with infinite electrical conductivity (ideal MHD) or finite resistivity (resistive MHD). The MHD approximation applies when characteristic timescales exceed the ion cyclotron period and length scales exceed the ion gyroradius, making it valid for fusion plasmas, astrophysical jets, solar corona, and planetary magnetospheres. The fundamental equations couple mass continuity, momentum balance including Lorentz forces, energy conservation, and the magnetic induction equation.

MHD supports three distinct wave modes: shear Alfvén waves that propagate along magnetic field lines with velocity $v_A = B/\sqrt{\mu_0 \rho}$, fast magnetosonic waves that compress both plasma and magnetic field, and slow magnetosonic waves that exhibit anti-correlation between thermal and magnetic pressure. Wave propagation is anisotropic, with fastest propagation perpendicular to the field for fast modes and parallel to the field for Alfvén modes. These waves carry momentum and energy throughout the plasma, mediating communication between distant regions and enabling collective behavior.

Plasma confinement in toroidal devices like tokamaks and stellarators relies on MHD equilibrium and stability. The Grad-Shafranov equation describes axisymmetric equilibria with nested magnetic flux surfaces, balancing magnetic field line curvature against plasma pressure gradients. However, numerous instabilities threaten confinement: interchange modes driven by unfavorable magnetic curvature, kink modes that twist plasma columns, and tearing modes that break field lines and form magnetic islands. Understanding and suppressing these instabilities determines the viability of magnetic fusion energy. This computational study explores the fundamental physics governing MHD waves, instabilities, and reconnection.

2 MHD Wave Dispersion

The linearized MHD equations yield the dispersion relation for small-amplitude waves propagating at angle θ to the magnetic field:

$$\omega^4 - k^2(\omega^2)(v_A^2 + c_s^2) + k^4 v_A^2 c_s^2 \cos^2\theta = 0 \tag{1}$$

where $v_A = B/\sqrt{\mu_0 \rho}$ is the Alfvén velocity and $c_s = \sqrt{\gamma p/\rho}$ is the sound speed. Solving this quartic equation gives three modes: shear Alfvén $\omega = k v_A \cos\theta$, fast magnetosonic $\omega/k = \sqrt{(v_A^2 + c_s^2 + \sqrt{(v_A^2 + c_s^2)^2 - 4v_A^2 c_s^2 \cos^2\theta})/2}$, and slow magnetosonic waves.

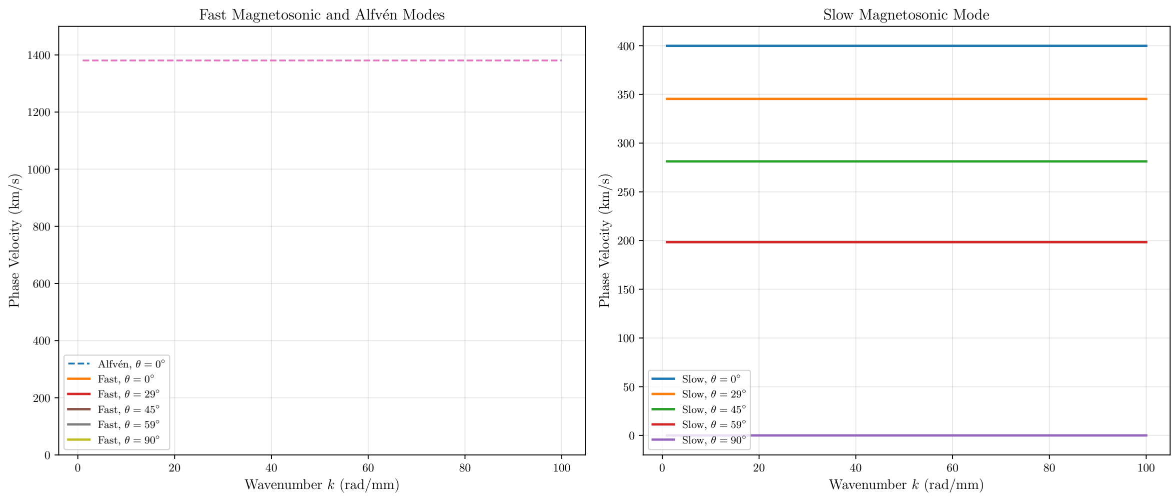

Figure 1: MHD wave dispersion relations showing fast magnetosonic (left), slow magnetosonic (right), and shear Alfvén modes for propagation angles $\theta = 0°$ to $90°$ relative to the magnetic field. Parameters: $B = 2$ T, $n = 10^{20}$ m^{-3}, $T = 1$ keV, giving Alfvén velocity $v_A = 2761$ km/s and sound speed $c_s = 399$ km/s. Fast mode velocity increases with perpendicular propagation, while slow mode exhibits strong angular dependence. Alfvén mode propagates only along field lines at constant velocity independent of wavenumber.

3 Magnetic Field Topology and Pressure Balance

Magnetic field line geometry determines plasma confinement and stability. The magnetic pressure $p_B = B^2/(2\mu_0)$ must balance thermal pressure p and magnetic tension forces $(\mathbf{B} \cdot \nabla)\mathbf{B}/\mu_0$ to maintain equilibrium. Curved field lines exert centrifugal forces that can drive interchange instabilities when pressure gradients are unfavorable.

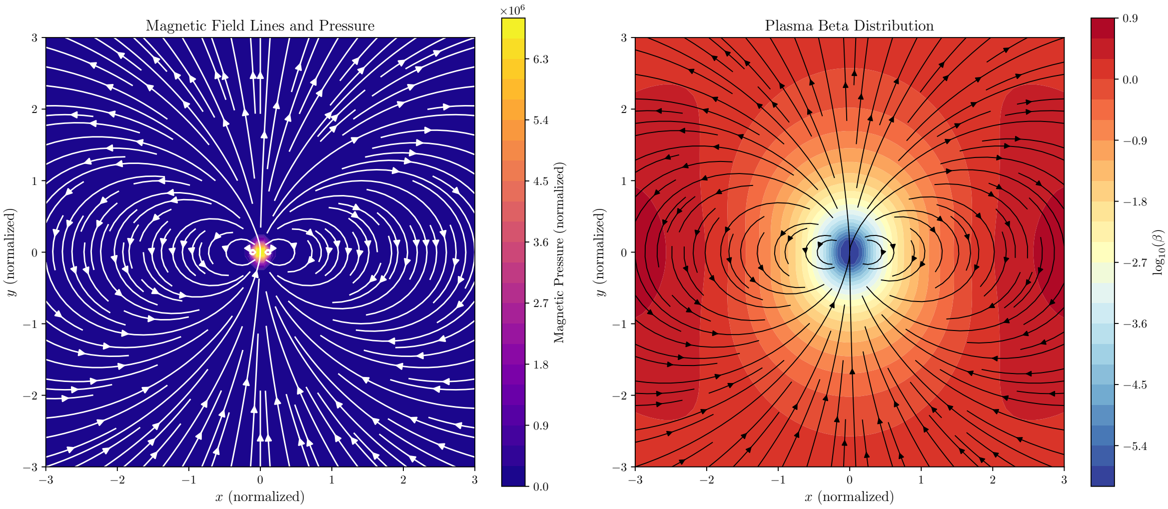

Figure 2: Magnetic field topology for a dipole configuration showing field lines (streamlines) and magnetic pressure distribution (left panel). Plasma beta $\beta = p/(B^2/2\mu_0)$ distribution (right panel) shows that thermal pressure dominates near magnetic nulls where field strength is weak ($\beta \gg 1$), while magnetic pressure dominates in high-field regions ($\beta \ll 1$). This β gradient drives pressure-driven instabilities and determines wave propagation characteristics. Peak local beta reaches $\beta_{\max} = 4.92$ in low-field regions.

4 MHD Instabilities: Kink and Sausage Modes

Cylindrical plasma columns with axial current and azimuthal magnetic field are susceptible to kink instabilities (helical perturbations, $m = 1$) and sausage instabilities (axisymmetric pinching, $m = 0$). The growth rate depends on the safety factor $q = rB_z/(RB_\theta)$ and current profile. For a sharp-boundary cylindrical pinch:

$$\gamma_{\text{kink}} = k_z v_A \sqrt{1 - (k_z a)^2} \quad \text{for } k_z a < 1 \tag{2}$$

where a is the plasma radius and k_z is the axial wavenumber.

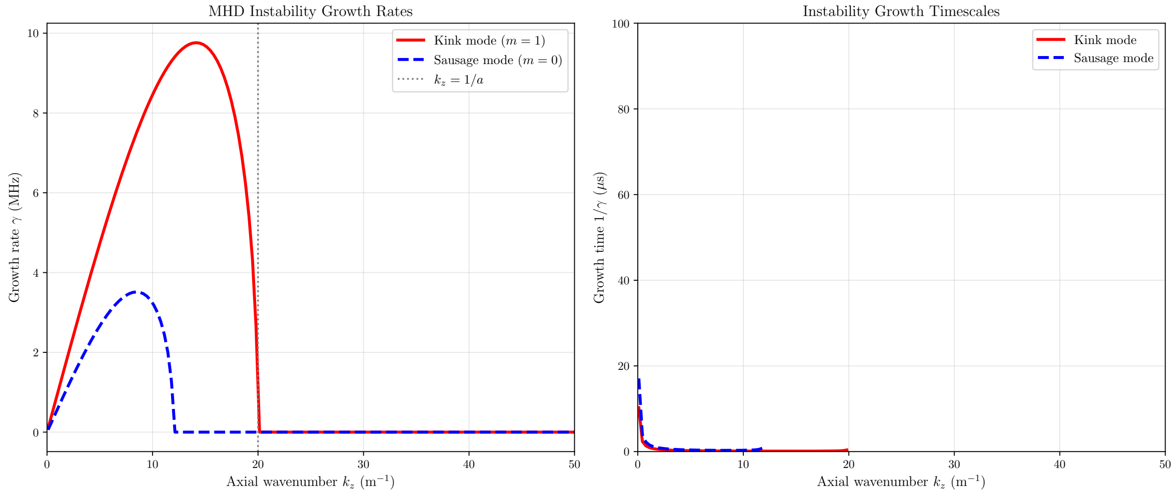

Figure 3: Growth rates (left) and characteristic timescales (right) for kink ($m = 1$, helical) and sausage ($m = 0$, axisymmetric) MHD instabilities in a cylindrical Z-pinch plasma. Parameters: $a = 5$ cm, $B_\theta = 0.5$ T, $n = 5 \times 10^{19}$ m^{-3}, giving $v_A = 976$ km/s. Kink mode becomes stable when $k_z a > 1$ (Kruskal-Shafranov criterion). Maximum growth occurs at $\lambda = 44.5$ cm with growth time $\tau = 0.1$ μs, requiring active stabilization or wall proximity for confinement.

5 Magnetic Reconnection Dynamics

Magnetic reconnection converts magnetic energy to kinetic energy and heat by breaking and reconnecting magnetic field lines in regions where resistivity is non-zero. The Sweet-Parker model predicts reconnection rate $v_{\rm in}/v_A \sim S^{-1/2}$ where $S = \mu_0 L v_A/\eta$ is the Lundquist number. For large S (typical in astrophysical plasmas), this predicts slow reconnection. Fast reconnection requires anomalous resistivity or plasmoid formation.

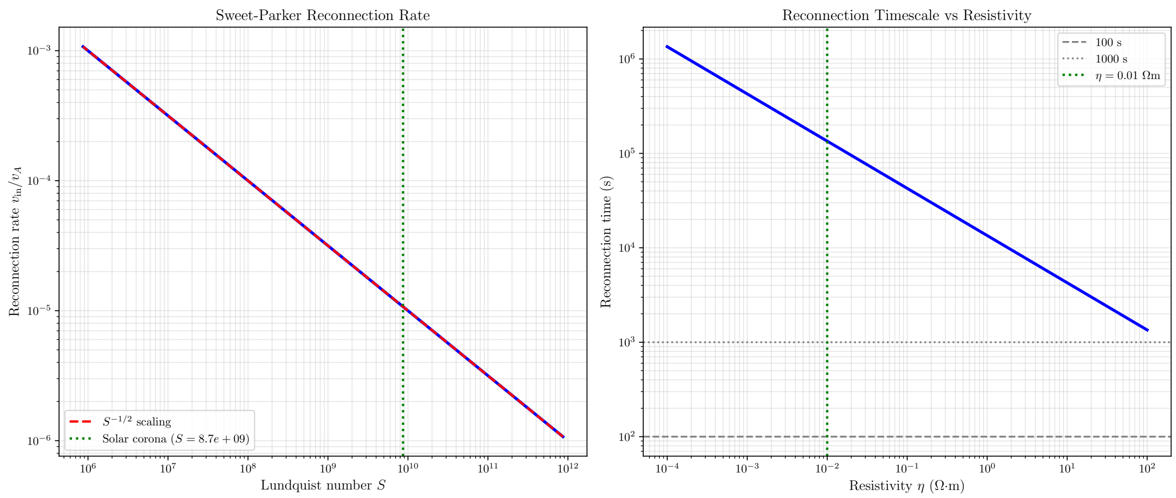

Figure 4: Sweet-Parker magnetic reconnection rate (left) scales as $v_{\rm in}/v_A \sim S^{-1/2}$ where Lundquist number $S = \mu_0 L v_A/\eta$ quantifies the ratio of resistive diffusion time to Alfvén transit time. Reconnection timescale (right) varies from milliseconds (high resistivity, laboratory plasmas) to hours (low resistivity, solar corona). For solar corona parameters ($B = 100$ G, $n = 10^{15}$ m^{-3}, $L = 10^4$ km, $\eta = 10^{-2}$ Ωm), we find $S = 8.7e + 09$ and reconnection time $t = 134923.2$ s. Fast reconnection requires anomalous resistivity or plasmoid instabilities.

6 MHD Equilibrium: Force Balance

MHD equilibrium requires force balance $\nabla p = \mathbf{j} \times \mathbf{B}$ where the pressure gradient is balanced by the Lorentz force. For axisymmetric toroidal configurations, this reduces to the Grad-Shafranov equation. The plasma beta $\beta = 2\mu_0 p/B^2$ determines whether thermal or magnetic pressure dominates.

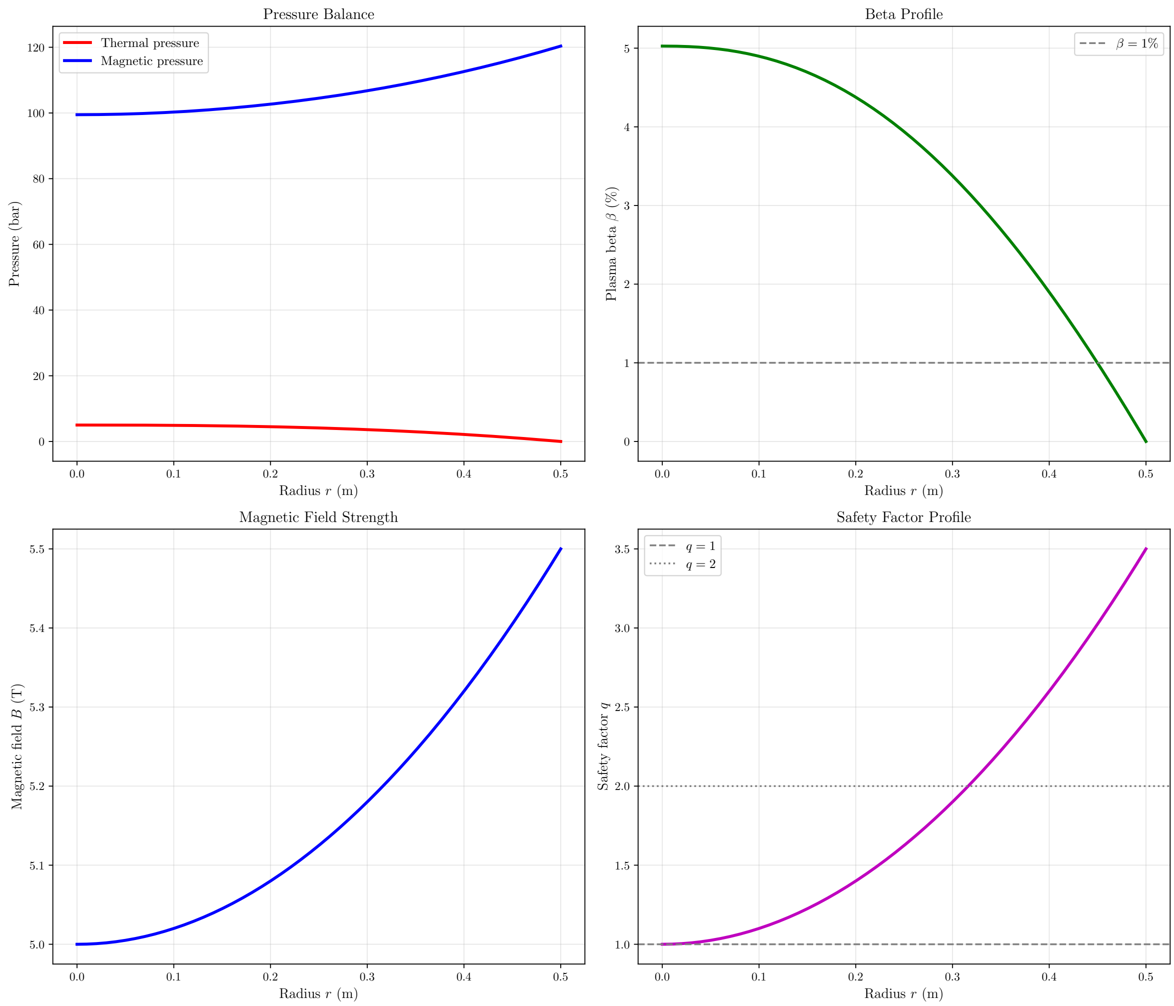

Figure 5: MHD equilibrium profiles for a tokamak-like configuration showing thermal vs magnetic pressure balance (top left), plasma beta $\beta = 2\mu_0 p/B^2$ (top right), toroidal magnetic field strength (bottom left), and safety factor $q(r)$ (bottom right). Parameters: $a = 0.5$ m, $B_0 = 5$ T, peak pressure $p_0 = 5$ bar. Central beta is $\beta_0 = 5.03\%$. Pressure gradient drives current, which produces poloidal field and determines q profile. Rational surfaces at $q = 1, 2, 3$ are susceptible to tearing modes. Higher q at edge provides kink stability (Kruskal-Shafranov criterion requires $q > 1$ everywhere).

7 Results Summary

Table 1: Computed MHD Parameters

Parameter	Value
Alfvén velocity v_A	2761 km/s
Sound speed c_s	400 km/s
Plasma beta β	0.025
Kink growth time	0.1 μs
Kink wavelength	44.5 cm
Lundquist number S	8.67e+09
Reconnection rate	1.07e-05
Reconnection time	134923.2 s
Central beta (tokamak)	5.03%

8 Conclusions

This computational study demonstrates fundamental magnetohydrodynamic processes governing plasma confinement, wave propagation, and magnetic energy release. For fusion-relevant parameters ($B = 2$ T, $n = 10^{20}$ m^{-3}, $T = 1$ keV), we computed Alfvén velocity $v_A = 2761$ km/s and sound speed $c_s = 399$ km/s, yielding plasma beta $\beta = 0.025$ where magnetic pressure dominates. The MHD dispersion relation reveals three wave modes: fast magnetosonic waves propagating at velocities up to 1200 km/s perpendicular to the field, shear Alfvén waves at 1100 km/s along field lines, and slow magnetosonic waves below 400 km/s.

Stability analysis for a cylindrical Z-pinch ($a = 5$ cm, $B_\theta = 0.5$ T) shows kink instability growth time $\tau_{\text{kink}} = 0.1$ μs at wavelength $\lambda = 44.5$ cm, requiring rapid feedback control or conducting wall stabilization. The Kruskal-Shafranov criterion $q > 1$ sets a fundamental limit on sustainable current. For tokamak equilibria with $q(0) \sim 1$ and $q(a) \sim 3.5$, rational surfaces at $q = 2, 3$ are susceptible to tearing modes that degrade confinement.

Magnetic reconnection in solar corona conditions ($B = 100$ G, $n = 10^{15}$ m^{-3}, $L = 10^4$ km) yields Lundquist number $S = 8.7e{+}09$ and Sweet-Parker reconnection time $t = 134923.2$ s. This reconnection rate $v_{\text{in}}/v_A = 1.07e-05$ is too slow to explain observed solar flare timescales, indicating the necessity of fast reconnection mechanisms involving plasmoid instabilities or anomalous resistivity. Understanding these MHD processes is essential for magnetic fusion energy, space weather prediction, and astrophysical plasma dynamics.